

EG1P12 Programming

Topic 3.1 Algorithm and Control
Statement

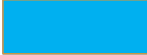
Learning Outcomes (LO)

1. Formulate real-world problems into computational algorithms using flowcharts.
2. Apply basic programming constructs to solve problems using computational thinking techniques.
3. Conduct testing and debugging on programs using appropriate techniques.
4. Develop structured computer programs to solve problems using computational thinking.

Flowchart (Topic 1.2)

- Structured steps help simplify processes.
- Flowcharts visually represent logic clearly
- A flowchart is visually representation of the steps involved in problem solving.
- Flowcharts uses symbols like arrows (flow), ovals (Start/End), parallelogram (Input/Output), diamond (decision) and rectangle (process)



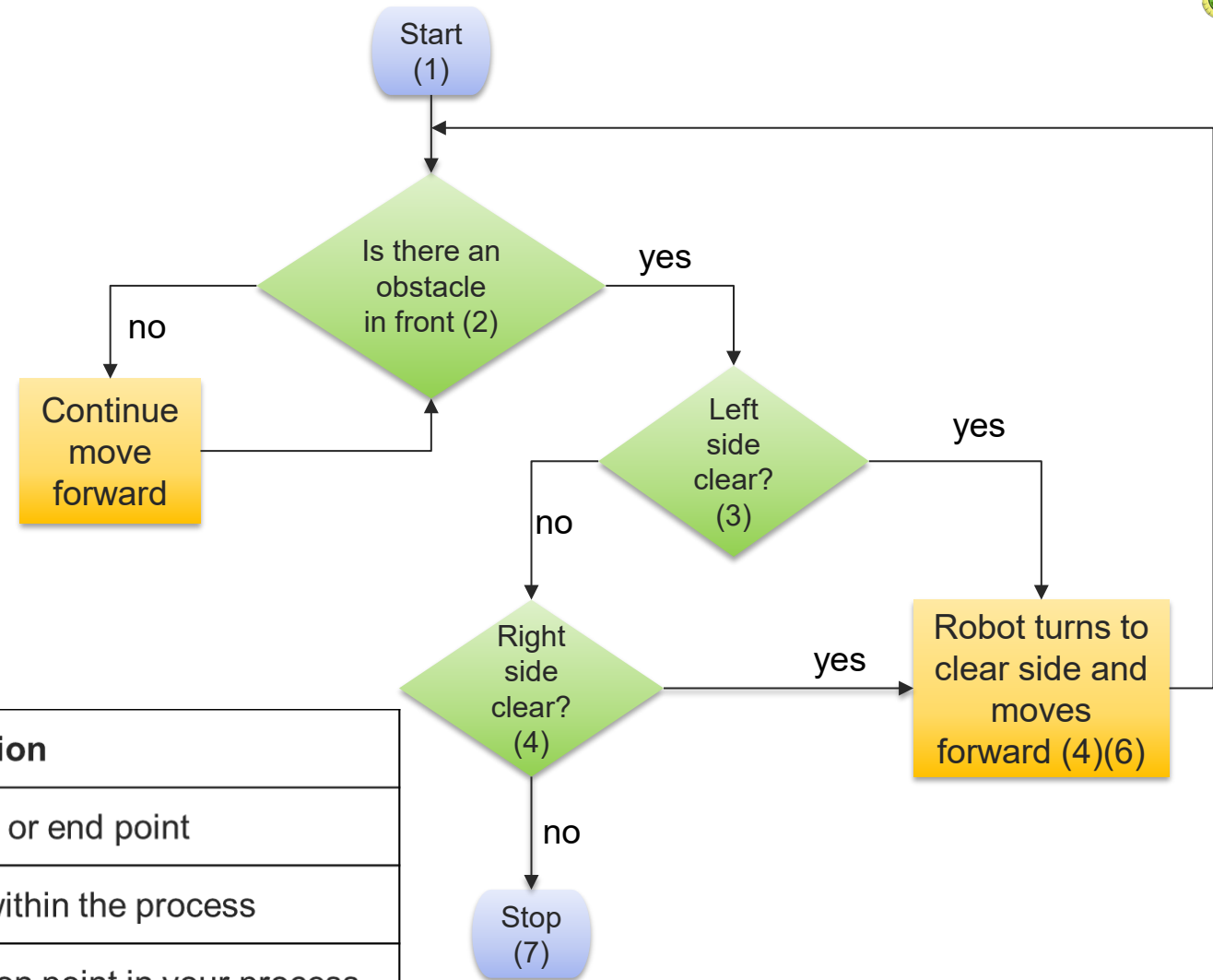
Symbol	Name	Function
	Terminal	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input / Output	A parallelogram represents input or output
	Processing	A rectangle represents a process
	Decision	A diamond indicates a decision

Flowchart (Topic 1.2)

Example: Robot Navigation

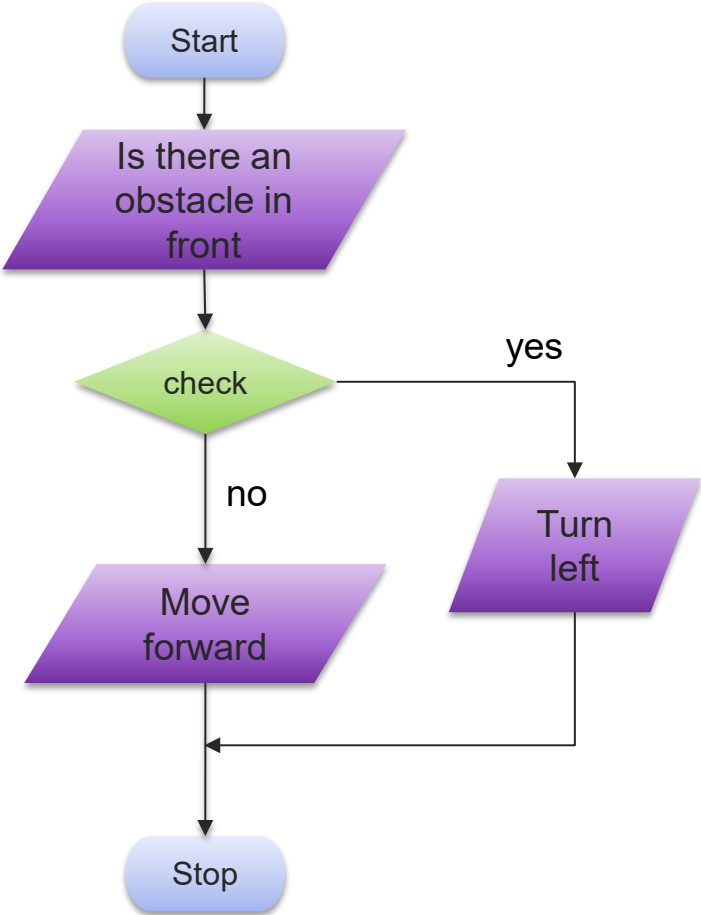
1. Start
2. Check for obstacles
3. If an obstacle is detected → Check left
4. If left is clear → Turn left
5. Otherwise → Check right
6. If right is clear → Turn right
7. Otherwise → Stop

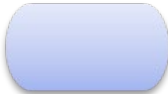
Symbol	Function
	A start or end point
	Step within the process
	Decision point in your process.
	Input or Output
	Flow



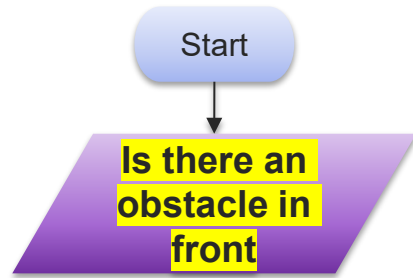


Decision Making – IF & Else statement



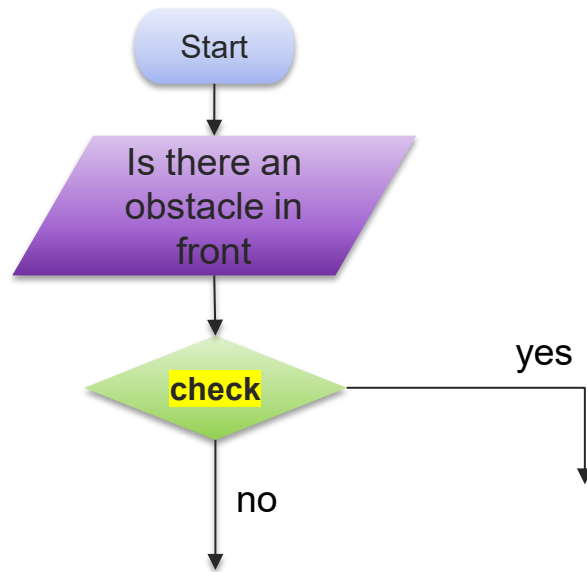
Symbol	Function
	A start or end point
	Step within the process
	Decision point in your process.
	Input or Output
	Flow

Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
```

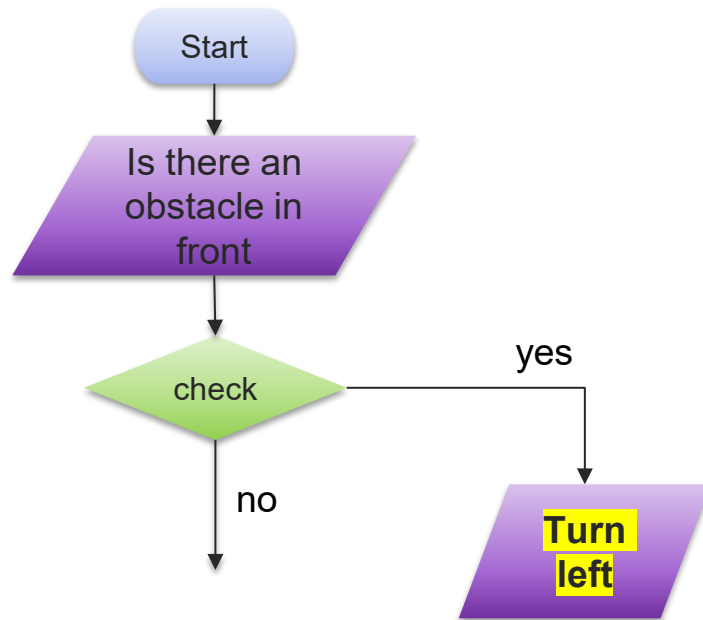
Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
```

Operator	Meaning	Example	Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	5 > 10	False
<	Less than	5 < 10	True
>=	Greater than or equal to	5 >= 5	True
<=	Less than or equal to	8 <= 5	False

Decision Making – IF & Else statement



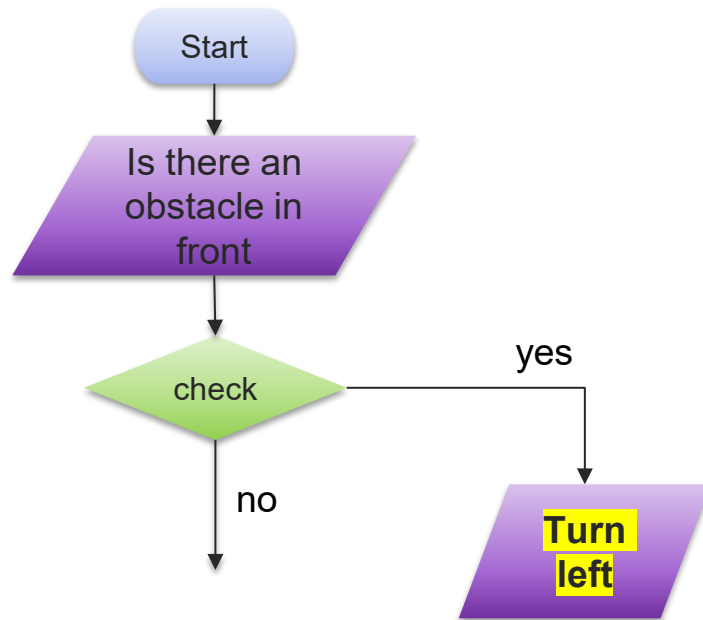
```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
```

Enter

```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Yes')
```

auto indentation (via Pycharm!!!)

Decision Making – IF & Else statement



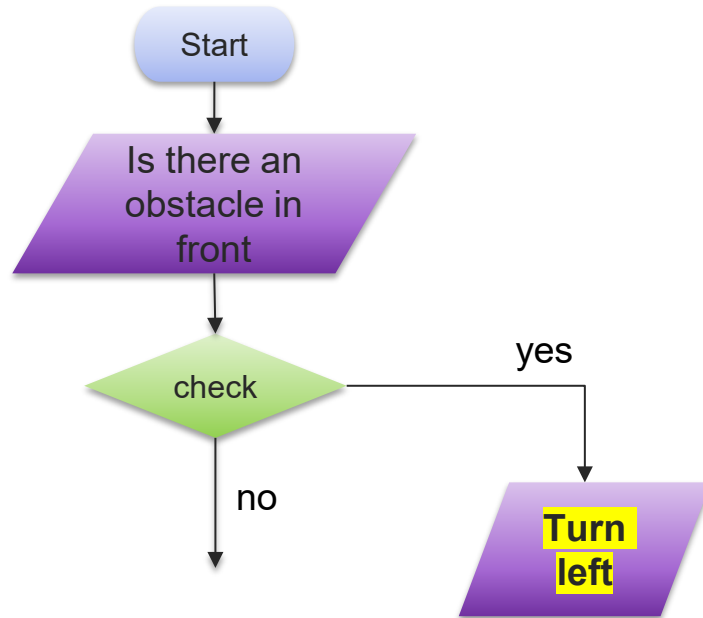
```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
```

```
1 obstacle = input('Is there an obstacle in front? yes/no: ').lower()
2 if obstacle == 'yes':
3     print('Yes')  Tab to complete
```

A. The "4-Space" Rule

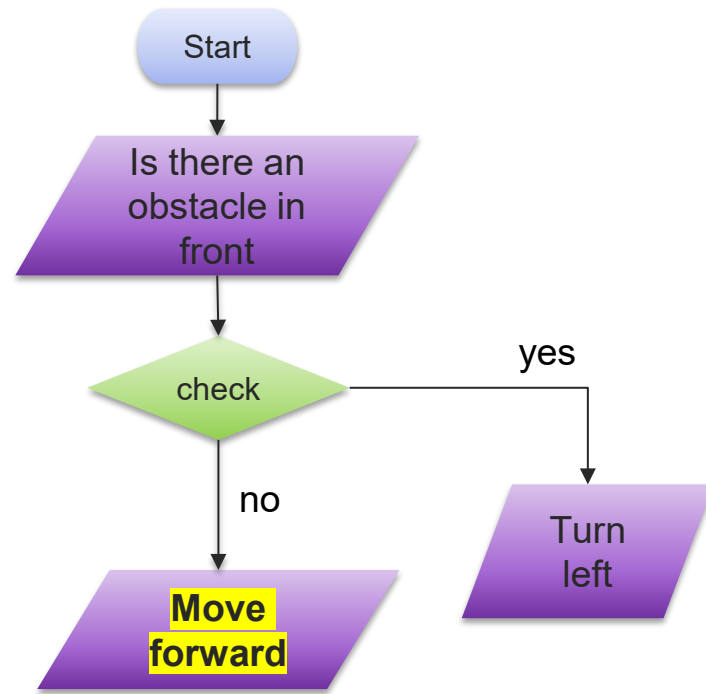
The official Python standard (PEP 8) recommends using **4 spaces** per indentation level. While you can use the `Tab` key, most professional IDEs are actually configured to turn one `Tab` press into four space characters.

Decision Making – IF & Else statement



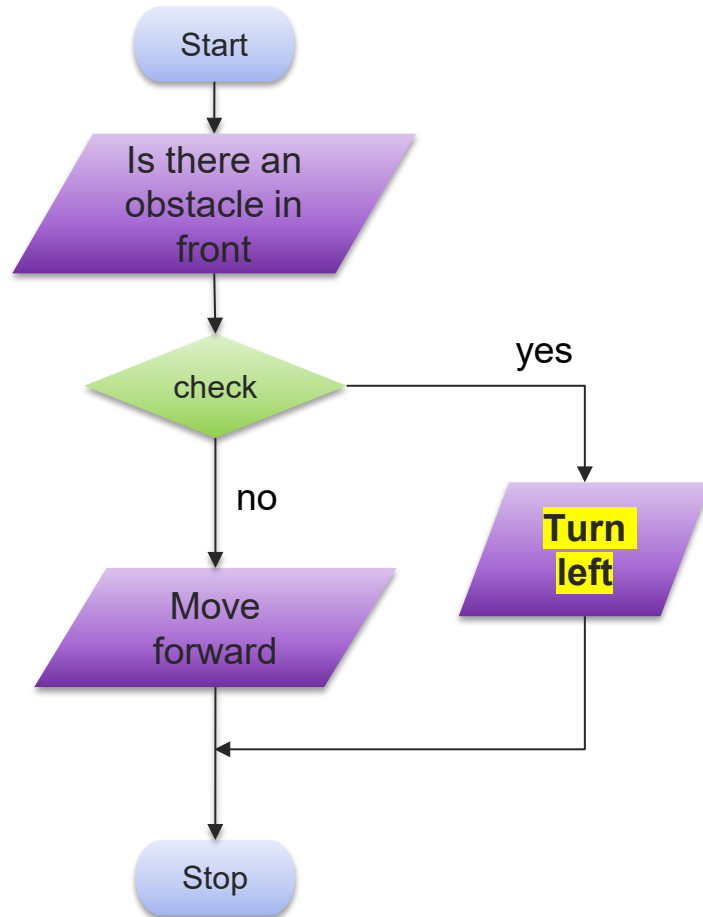
```
1  obstacle = input('Is there an obstacle in front? yes/no: ')
2  if obstacle == 'yes':
3      print('Turn left')
```

Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

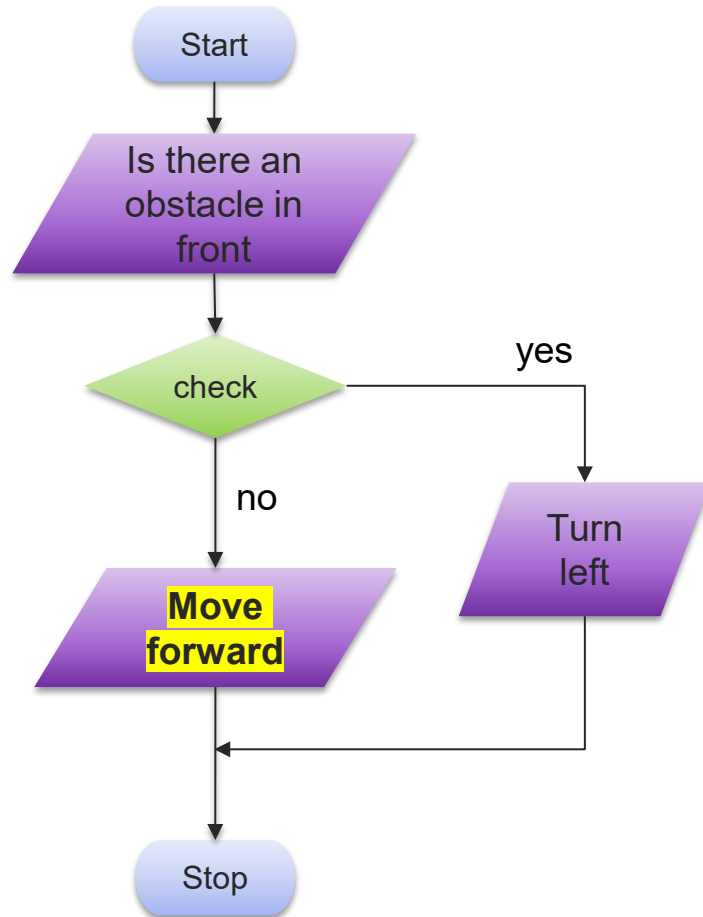
Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

```
Is there an obstacle in front? yes/no: yes
Turn left ←--
Process finished with exit code 0
```

Decision Making – IF & Else statement

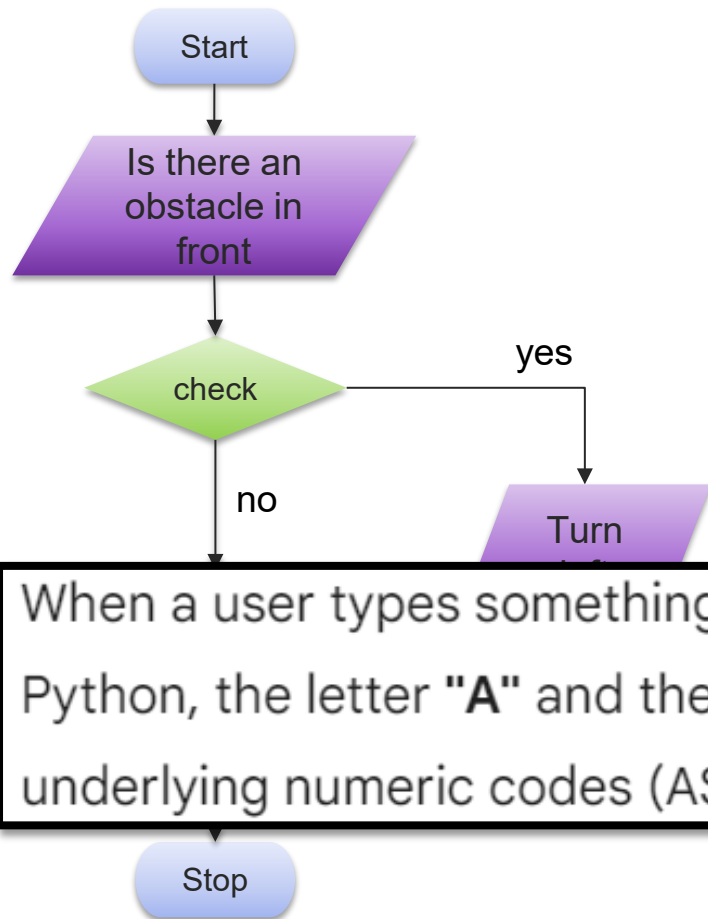


```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

```
Is there an obstacle in front? yes/no: YES
Move forward <---
Process finished with exit code 0
```

WHY?

Decision Making – IF & Else statement

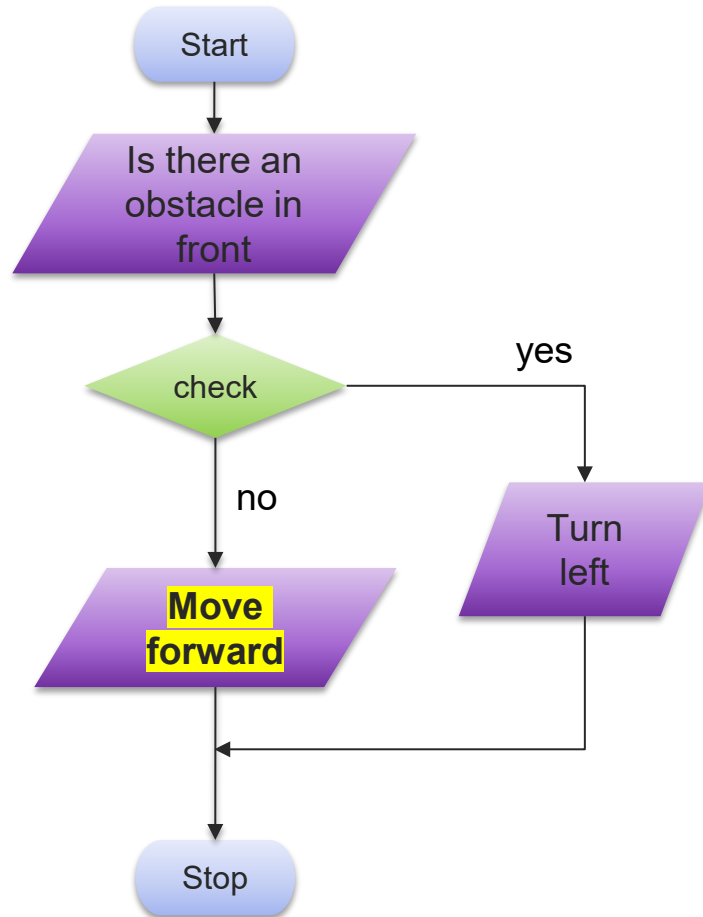


```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

When a user types something into the console, Python captures it exactly as it was typed. To Python, the letter **"A"** and the letter **"a"** are **completely different characters** with different underlying numeric codes (ASCII/Unicode).

WHY?

Decision Making – IF & Else statement

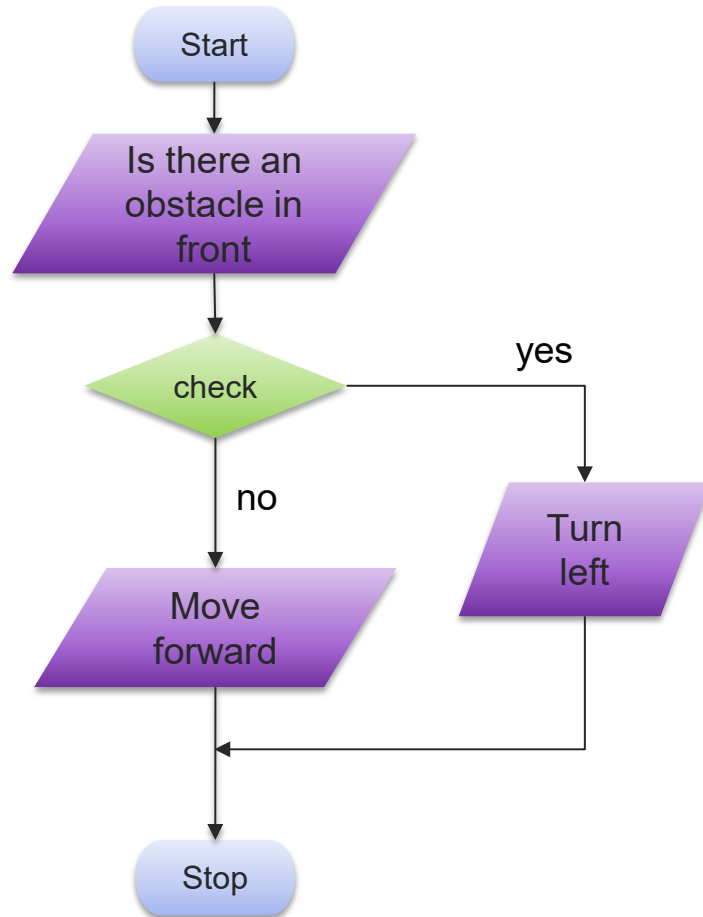


```
1 obstacle = input('Is there an obstacle in front? yes/no: ')
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

```
Is there an obstacle in front? yes/no: YES
Move forward <---
Process finished with exit code 0
```

How to resolve?

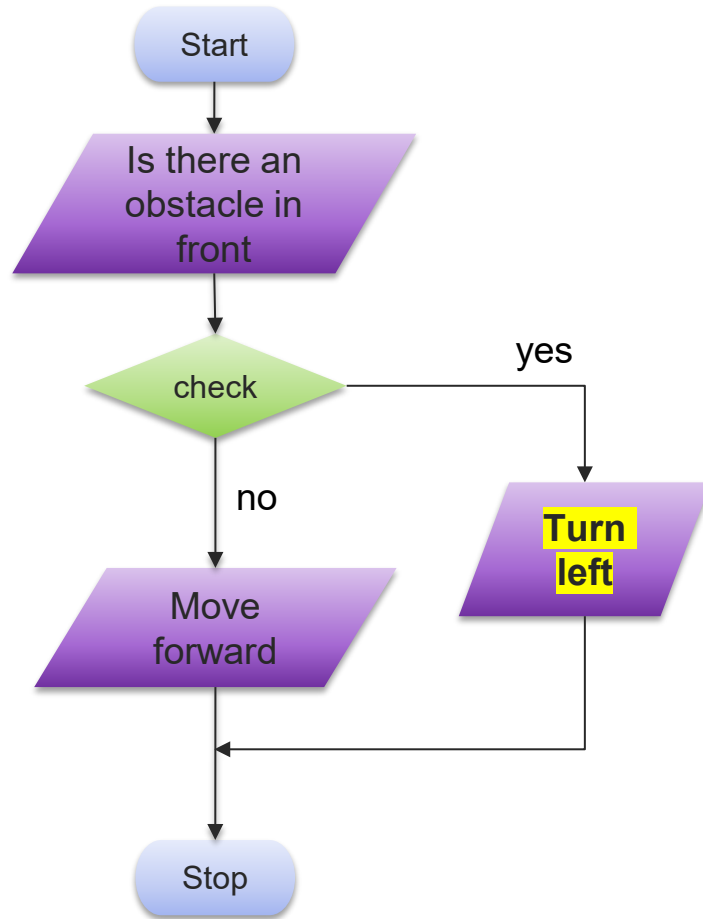
Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ').lower()
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

.lower() converts whatever the user types into lowercase.

Decision Making – IF & Else statement

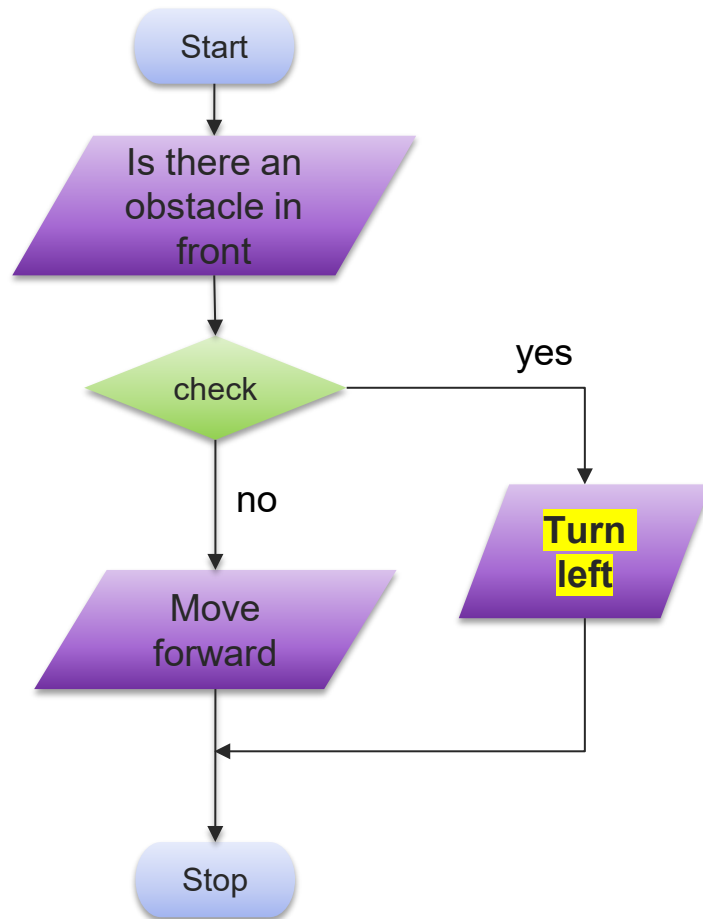


```
1 obstacle = input('Is there an obstacle in front? yes/no: ').lower()
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')]
```

```
Is there an obstacle in front? yes/no: YES
Turn left

Process finished with exit code 0
```

Decision Making – IF & Else statement



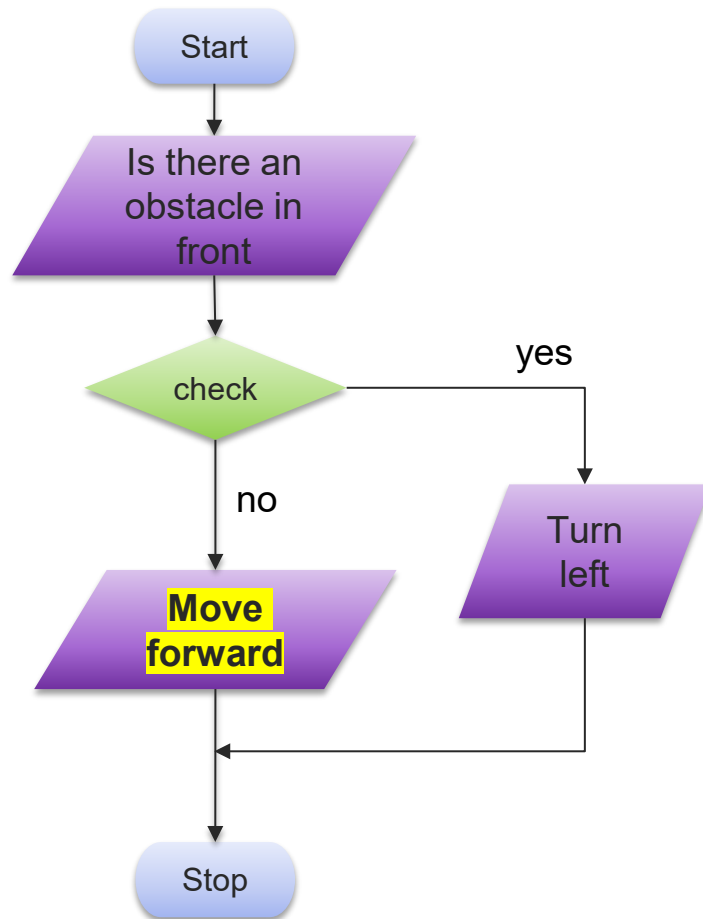
```
1 obstacle = input('Is there an obstacle in front? yes/no: ').lower()
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')
```

```
Is there an obstacle in front? yes/no: YES
Turn left

Process finished with exit code 0
```

if and else must be aligned

Decision Making – IF & Else statement



```
1 obstacle = input('Is there an obstacle in front? yes/no: ').lower()
2 if obstacle == 'yes':
3     print('Turn left')
4 else:
5     print('Move forward')]
```

```
Is there an obstacle in front? yes/no: YES
Turn left

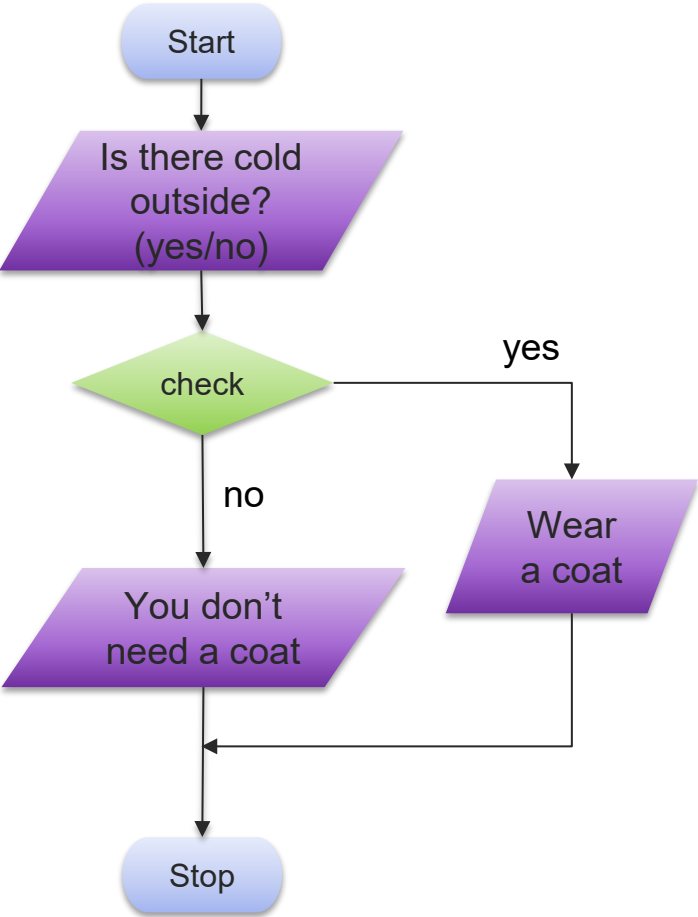
Process finished with exit code 0
```

What will happened if you input any other character(s), other than yes/no?



Decision Making – IF & Else statement

Identifying issue(s)



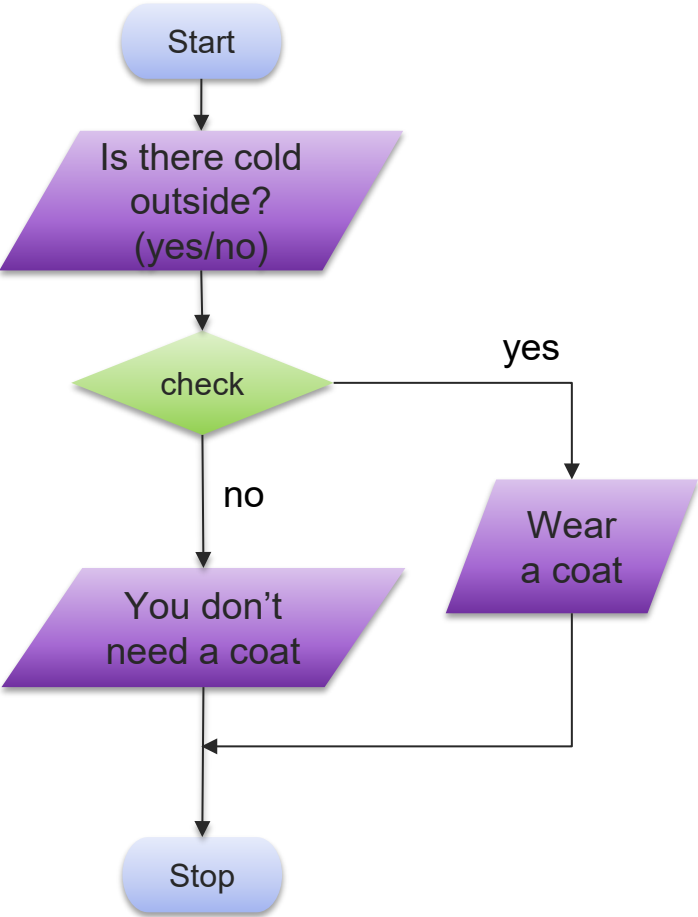
```
1 temp = input('Is it cold outside? (yes/no): ')
2 if temp == 'yes':
3     print('wear a coat')
4 else:
5     print("You don't need a coat")
```

Any issue?



Decision Making – IF & Else statement

Identifying issue(s)



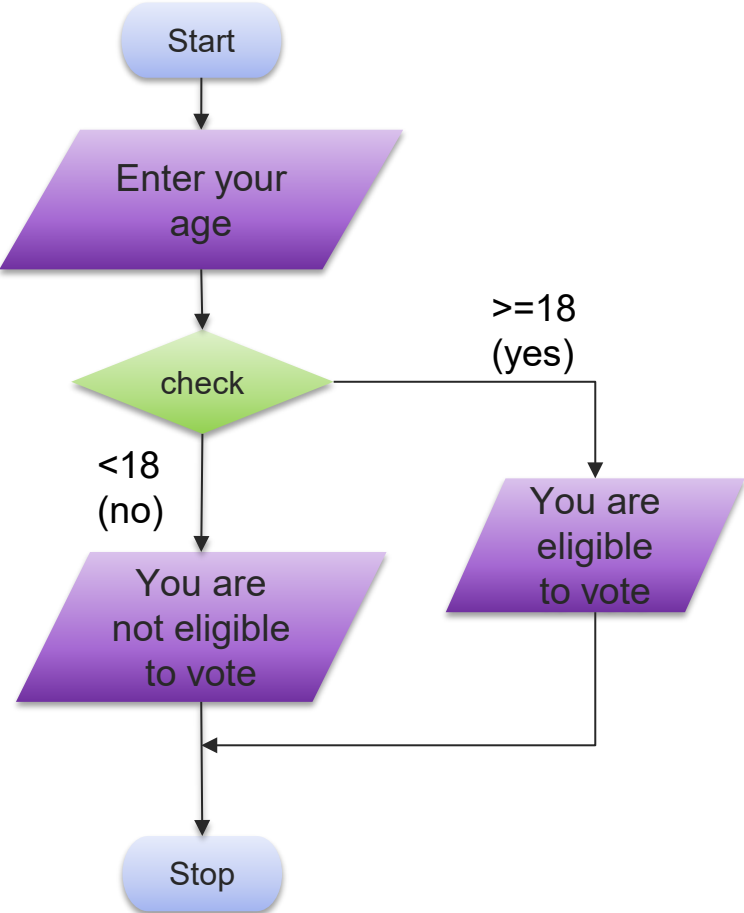
```
1 temp = input('Is it cold outside? (yes/no): ')
2 if temp == 'yes':
3     print('wear a coat')
4 else:
5     print("You don't need a coat")
```

Indentation is impt



Decision Making – IF & Else statement

Example – Voting Eligibility



```
1 ageCheck = int(input("Enter your age: "))
2 if ageCheck >= 18:
3     print("You are eligible to vote")
4 else:
5     print("You are not eligible to vote")
```

```
Enter your age: 18
You are eligible to vote
Process finished with exit code 0
```

```
Enter your age: 10
You are not eligible to vote
Process finished with exit code 0
```

Thank You!



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